

Self-Fulfilling Forecasts in a Minimal LLM Market: An Experimental Test of Performative Feedback Strength

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Abstract

Predictions can influence the outcome of the market they are meant to predict when they become an input to the process leading to the later outcome. This phenomenon is called “performativity.” Several theoretical studies suggest that increased feedback between predictions and outcomes in a multi-agent prediction system, under idealized learning-agent dynamics, can increase instability and move the system from stable to chaotic dynamics. It is still unclear whether this type of instability can also arise in a multi-agent prediction system made up of realistic and unconstrained agents, such as Large Language Models (LLMs).

To investigate this empirically, we designed a minimal multi-agent LLM market based on an established LLM market model. The realized price in this market is a convex combination of the fundamental value and the agents’ predictions, with the parameter ε determining the importance of the fundamental value. We used $\varepsilon \in \{0, 0.2, 0.4, 0.6, 0.8\}$, eight different initial conditions (seeds), and forty rounds. The price differences were decomposed into a mechanical component coming from the price formation rule and a residual amplification component.

We did not find a statistically significant effect of feedback intensity on whether agents deviated from the mechanical baseline in their predictions ($p = 0.903$). The main change we observed with feedback intensity was in the magnitude of these deviations. As feedback intensity increased, both the absolute size and the variance of the deviations increased. The relationship was strong ($R^2 = 0.833$) and was tested using OLS regression with heteroskedasticity-robust standard errors and a distribution-free Spearman rank correlation. At first, we expected reasoners using an extrapolative strategy, who tend to extend the current trend into the future, to be more sensitive to changes in feedback intensity. This was not what we found. Agents with corrective personalities were more sensitive to changes in feedback intensity.

Our findings suggest that increased performativity can be observed in a market consisting of LLM agents, even when these agents do not follow the learning dynamics assumed by the dominant theory. The effect is not the same across all agents. Instead, it seems to increase the heterogeneity in their behavior. Therefore, empirical models of instability in performative systems should account for growing behavioral heterogeneity alongside dynamic chaos.

1 Introduction

Any publicly disseminated forecast can influence the behavior that the forecast tries to predict, according to Morgenstern (1928). Keynes (1936) explained a similar idea via his beauty-contest analogy of asset pricing, while the same idea was articulated by Soros (2013) as “reflexivity.” Hofstadter (1979) referred to the same phenomenon as a “strange loop.” Perdomo et al. (2020) provided a formalized treatment of this idea in the machine learning context, calling it “performative prediction.” Later, Piliouras and Yu (2023) formalized it in a multi-agent setting, showing that increasing feedback strength can take an idealized forecasting mechanism from a regime of stability to a regime of formal (Li–Yorke) chaos. Their theoretical result establishes a benchmark for studying

the effects of increased performativity on the stability of forecasting systems. Note that their analysis is based on idealized gradient-based learning agents.

There is a second literature on the subject that shows empirically that LLM agents interacting in a live market setting, where the actions of agents result in price changes, have distinct dynamics under fixed positive and negative feedback regimes. See, for example, del Rio-Chanona et al. (2025). It should be noted, however, that the feedback strength was not varied in this work. Kleitsikas et al. (2025) introduced a continuous performativity parameter into a financial market-making model using closed-form stochastic control and a single analytical strategy rather than learning agents.

To the best of our knowledge, there has been no empirical investigation of how a continuously varying feedback strength parameter affects the price deviations induced by forecasts when the forecasting agents are real LLMs rather than idealized learners, in particular. In our study, we aim to bridge this gap in the literature. For that, we use the LLM-market pipeline, which was validated by del Rio-Chanona et al., as an experimental setup. We use the stability results of Piliouras and Yu (2023) as a theoretical benchmark for analyzing the effect of varying feedback strength.

1.1 Our Contributions

Our paper makes three major contributions. Firstly, we propose a continuous feedback-strength parameter, ε , defining price formation in the repeated forecasting game of LLM agents. With this introduction, we manage to unite in one common parameter, that can be varied continuously, two static regimes of feedback used in previous experiments with LLM-market simulations. Thus, the effect of the increase in performativity can be analyzed systematically.

Secondly, as the realized price is a convex combination of the fundamental value and agents' aggregate forecast, we can suggest a mechanical-amplification decomposition of deviations. The idea is to separate a mechanical deviation, created by the price formation rule itself, from an additional amplification, made on top of that base. It is necessary, as in a straightforward analysis of the effect of performativity increase on deviations, we may get a positive correlation mechanically, since the feedback strength parameter ε influences directly the weight of the forecast.

Finally, we analyze the dynamics created by the described mechanism in two ways: aggregated and persona-by-persona. We do not observe any stable tendency towards the growth or decrease of the deviations caused by forecasting in case of increase in ε . On the contrary, we see greater dispersion and higher absolute deviations from the fundamental value for the increasing performativity level. Additionally, we test a previously suggested hypothesis at the persona level — that extrapolative reasoning styles would show greater feedback sensitivity than corrective ones — which was not supported. We think that this result is more informative than the absence of the hypothesis confirmation itself.

2 Related Work

2.1 Performative Prediction Theory

Perdomo et al. (2020) formally introduce the concept of performative prediction in the case of a single learning agent. This theory includes notions of performative optimality and performative stability, and the analysis of the framework in which the predictions made by a model can impact the distribution of data on which the future predictions of the model are evaluated. This feature of the framework becomes especially relevant in the case of multiple learning agents. Piliouras and Yu (2023) extend the framework in this direction and prove that the increase in collective influence may change the system dynamics from stable to formal Li-Yorke chaos. However, their

theoretical analysis is based on the ideal case of gradient-based learning agents. It therefore provides a theoretical benchmark for our study. The question here is whether stronger feedback has a similar effect on the forecasting system when using real LLMs. For further reading on the topic, refer to Kehrenberg et al. (2026).

2.2 Reflexivity and Self-Referential Forecasting in Economics

Inconsistency between the predictions and the behavior generated by them has had many different manifestations in economics and other fields. The problem of the inconsistency of predictions and the behavior created by them originated in economics itself. It was discussed by Morgenstern (1928), who noted that a prediction made by an authoritative person might affect the subject’s behavior that should be predicted. The problem of reflexivity in economics was studied in a fixed-point setting by Grunberg and Modigliani (1954), who showed that self-referentiality does not prevent economic forecasting. Keynes (1936)’s beauty contest game for asset pricing serves as yet another example of a self-reflexive situation, where agents try to make predictions based on the expectations of other agents. A similar phenomenon of feedback from perceptions to reality was addressed in the framework of the notion of reflexivity by Soros (2013). He suggested that the perceptions of the state by the agents could affect the state itself, and changes in the state could also affect agents’ perceptions. Hofstadter (1979) also studied a similar case of recursion from cognitive and philosophical perspectives. He coined the term “strange loop” for it. All these examples of scientific work have little in common except for one thing — they discuss situations where representations or expectations of the system in question become an integral part of the changing system. This framework will be used in the computational marketplace environment in which predictions from LLMs are incorporated into the price formation process.

2.3 LLM-Agent Market Simulation and Discrete Feedback Structures

Recent work has started to examine whether large language models can produce economically meaningful behavior as market participants. del Rio-Chanona et al. (2025) use the classical experimental market framework of Heemeijer et al. (2009) with LLM agents. They find that both GPT-3.5 and GPT-4 show boundedly rational forecasting behavior rather than rational expectations under fixed positive and negative feedback price rules. Another example is TwinMarket (Yang et al. 2025), which shows that large-scale market simulations with multiple LLM agents are possible. Rather than holding feedback structures constant to compare positive and negative feedback—as is standard in the literature—we investigate how market behavior responds directly to fluctuations in performative strength.

2.4 Continuous Performativity Parameters

Kleitsikas et al. (2025) proposed a financial market-making model that includes a continuously changing performative-sensitivity parameter. Their model allows the effects of performativity in financial markets to be studied using stochastic control techniques. However, their main focus is on finding an analytical solution to the problem, while our study uses a different approach — real, unconstrained LLM agents rather than a closed-form analytical strategy.

There is still not much evidence on how LLM agents behave when the strength of performativity changes. Our study therefore looks at how a changing feedback parameter affects a heterogeneous group of LLM forecasters. In particular, we examine whether stronger performativity changes the direction, magnitude, and dispersion of deviations.

3 Framework and Experimental Design

3.1 Price Formation Rule

The price formation rule is a combination of a benchmark and a sum of the forecasts of the forecasting agents in the population. Price realization in each round t can be described as

$$P_t = (1 - \varepsilon)F_t + \varepsilon \text{Agg}(f_{1,t-1}, \dots, f_{5,t-1}),$$

where F_t is the fundamental value at time t , $f_{i,t-1}$ is the forecast of agent i in the previous round, and $\varepsilon \in [0, 1]$ is a parameter defining the performative feedback level in the market.

In contrast to constant positive- and negative-feedback price rules (del Rio-Chanona et al. 2025), used in previous LLM-agent market experiments, this framework allows for modeling the continuum of feedback levels with a single parameter. In particular, for $\varepsilon = 0$ the price equals the fundamental benchmark and does not include performative feedback, while for $\varepsilon = 1$ the price equals the aggregate of agents’ forecasts,

$$P_t = \text{Agg}(f_{1,t-1}, \dots, f_{5,t-1}).$$

3.2 Fundamental Process

The fundamental process is modeled by a geometric random walk:

$$F_t = F_{t-1} \exp(\mu + \sigma z_t), \quad z_t \sim \mathcal{N}(0, 1),$$

with initial value $F_0 = 100$, $\mu = 0.001$, $\sigma = 0.02$. For these parameter values, the fundamental process over the 40-round experimental horizon shows a non-trivial process with noisy dynamics: there is on average a 4% cumulative log-drift and 12.6% cumulative log-volatility.

3.3 Agent Population

The market environment features five LLM forecasting agents, each with one of five forecasting personas: trend-following, mean-reverting, fundamentals-anchored, momentum-sensitive, and neutral. Such heterogeneity is justified by the necessity of adding variability to the agents’ forecasting behavior so that no homogeneous forecasts-based dynamics would emerge, which is a potential threat identified by del Rio-Chanona et al. (2025). Their work showed that LLM agents are less heterogeneous in their behavior compared to human agents.

Two of the five forecasting personas — mean-reverting and fundamentals-anchored — were created taking into account not just the direction of correction, but also the time horizon. While the mean-reverting persona responds to the short-run average calculated for the last few rounds, the fundamentals-anchored persona responds to the long-run average calculated throughout the entire period of history, particularly considering the absence of the short-run momentum effect.

This was a necessary decision to make due to the absence of the fundamental value F_t (Section 3.4) and the possibility of implementing the instruction “anchor on the long-run value” through the average price over observed periods only, which would otherwise result in duplication of the mean-reverting persona.

The personas remained constant across the experimental conditions, enabling us to analyze their effects on the market outcomes across the ε levels.

3.4 Information Visibility

The price history $P_1, \dots, P_{t-1}$ is all the information the agents have access to when making their forecasts, as the contemporaneous fundamental value F_t is never observed. This visibility of information corresponds to the contrast between observable market prices and the unobservable underlying process of fundamentals, and can be seen in both the human laboratory experiment of Heemeijer et al. (2009) as well as its LLM-agent version used by del Rio-Chanona et al. (2025). Importantly, this limitation to price history allows for the retention of the meaning of the mechanical-amplification decomposition described later. If the agents were to have access to F_t , they could detect any deviations caused by the feedback mechanism and correct for them, confusing the mechanical deviation from prices with amplification from the forecasting process.

3.5 Mechanical-Amplification Decomposition

A higher ε mechanically places more weight on the agents' aggregate forecasts in the price formation process. Hence, the existence of larger deviations of prices from fundamentals does not in itself imply a change in the behavior of the agents' forecasts as a result of a change in ε ; part of this deviation is explained by the fact that the deviations are generated by the mechanics of the price rule itself.

In order to distinguish between the two effects, for each seed we calculate the bias in the agents' forecasts in the non-performative environment, $\varepsilon = 0$:

$$B_t(0) = \text{Agg}(f_{1,t-1}^{(0)}, \dots, f_{5,t-1}^{(0)}) - F_t,$$

where the superscript (0) indicates that we use the forecasts collected for $\varepsilon = 0$. The observed deviation is denoted as $D_t(\varepsilon) = P_t(\varepsilon) - F_t$ and the mechanically expected deviation at the level of feedback ε is equal to:

$$D_t^{\text{mech}}(\varepsilon) = \varepsilon \cdot B_t(0).$$

The residual of the amplified deviation is calculated as:

$$D_t^{\text{amp}}(\varepsilon) = D_t(\varepsilon) - D_t^{\text{mech}}(\varepsilon).$$

We interpret $D_t^{\text{amp}}(\varepsilon)$ as an amplification not explained by the mechanically expected effect of the feedback rule. Hence, this decomposition identifies two possible sources of price deviation: (i) the direct mechanical effect of increasing the weight on the agents' forecasts, and (ii) amplified price deviation due to changes in agents' forecasting behavior as a result of a performative market environment.

3.6 Implementation and Reproducibility

The predictions were generated using gpt-5.6-luna through the OpenAI-compatible chat completions API. Temperature was not set, so the default value was used. This is a limitation, but the eight seeds partly compensate for it, since F_t is fixed within each seed and the same trajectory is followed across all five ε values. Each agent received a persona-specific system prompt and the price history, and was asked to return a JSON object with reasoning and a numerical prediction. The API's structured output feature was not used; the text response was parsed to JSON instead. On failure, the call was retried up to two times; if it still failed, the previous round's price was used as the prediction. The five persona predictions were then averaged with a simple unweighted

average, $\text{Agg}(\cdot)$, as in Eq. (1). The experiment covered $\varepsilon \in \{0, 0.2, 0.4, 0.6, 0.8\}$, 8 seeds, and 40 rounds — 1600 seed- ε -round cells in total, with the five persona predictions requested at the same time within each round.

4 Results

The performativity measure (ε) fails to generate any clear directional changes in the trend of the average deviation of the price from its fundamental value. On the other hand, as the performativity effect increases, the size and volatility of price deviations become greater. Our analysis starts with the mechanical base (Section 4.1), followed by the directional amplification test (Section 4.2). We then consider the size and volatility of deviations (Section 4.3), after which we discuss persona-specific results (Section 4.4). The round-wise analysis of price dynamics is discussed in Section 4.5.

4.1 Mechanical Baseline

For each seed, the value of forecast bias $B_t(0)$ was calculated assuming there is no performativity effect ($\varepsilon = 0$). This value was used as a baseline to calculate the mechanically expected price deviation $D_t^{\text{mech}}(\varepsilon) = \varepsilon \cdot B_t(0)$ for every value of ε , as described in Section 3.5.

4.2 Directional Amplification (H1b)

For testing the hypothesis H1b, we checked if there is any systematic change in the residual deviation $D_t^{\text{amp}}(\varepsilon)$ with respect to ε . The regression was conducted with 32 observations of the seed-level data (8 seeds $\times$ 4 non-zero ε levels), excluding the case when $\varepsilon = 0$ since this is the basic situation without performative feedback. We estimate $\beta = 0.084$, which remains statistically indistinguishable from zero ($p = 0.903$, HC3-robust). The coefficient of determination is likewise negligible ($R^2 = 0.001$), confirming that ε explains practically none of the variance in residual deviations.

Each of the eight seeds provides one data point per ε value, yielding a total of 32 data points for the seeds-by- ε combination that is not entirely independent. HC3 does not solve this issue of correlation within seeds. Therefore, the regression was estimated again using three other methods: clustering of standard errors by seed, seed fixed effects, and the exact wild cluster bootstrap (Cameron et al. 2008). In the latter case, all $2^8 = 256$ combinations of signs across eight seeds are tested without using the asymptotic approximations. All three methods provided similar conclusions to the original analysis, i.e., that the ε coefficient is still insignificant ($p = 0.930$ for clustered standard errors, $p = 0.939$ for seed fixed effects, $p = 0.930$ for exact bootstrap).

These results fail to support hypothesis H1b. Scaling ε up does not push aggregate forecasts systematically above or below the fundamental value, meaning the added deviation lacks a consistent direction across seeds, and this conclusion is robust to accounting for the repeated-seed structure of the data.

4.3 Volatility and Magnitude of Amplification (H2)

Examining the spread of these deviations reveals a markedly different dynamic. While ε leaves the direction of deviations untouched, it strongly drives overall volatility. To calculate volatility, we estimated the standard deviation of deviation for each seed- ε combination. As Figure 1 shows, volatility increases along with the increase of ε ($\beta = 3.14$, $p < 0.001$, $R^2 = 0.833$). The same holds even when HC3 heteroskedasticity-consistent standard errors are applied to the regression

($p < 0.001$). Finally, our results are confirmed by the Spearman rank correlation ($\rho = 0.966$, $p < 0.001$).

Similar tendencies can be observed when considering the size of residual amplification. The absolute value of $D_t^{\text{amp}}(\varepsilon)$ increases with the increase of ε (OLS: $\beta = 1.86$, $R^2 = 0.338$, $p < 0.001$). It still holds true under the robust standard errors estimation method (HC3) ($p = 0.003$), and there is a strong positive correlation according to the Spearman test ($\rho = 0.817$, $p < 0.001$).

We tested the volatility and amplification magnitude results against the same set of three robustness tests. In terms of volatility, the result is statistically significant in the case of seed-clustered standard errors ($p < 0.001$), seed fixed effects ($p < 0.001$), and the exact wild cluster bootstrap ($p = 0.004$). With regard to the residual amplification magnitude, the result is statistically significant regardless of the specification, namely: $p = 0.004$ when using clustering; $p = 0.011$ in the case of fixed effects; and $p = 0.008$ when applying the exact bootstrap test. Thus, all three approaches support the consistency of our results concerning volatility and amplification.

Overall, the results above confirm H2. With an increase in performative feedback, forecasting deviations do not move in any particular direction. Instead, they become greater and more volatile. The bias can be positive for some seeds and negative for others; however, the absolute value of the bias increases.

4.4 Persona-Level Analysis

The difference in sensitivity to ε across persona types was noted earlier. We expected extrapolative personas to be more sensitive to ε because of their tendency to follow momentum. However, the results show that corrective personas are more sensitive to ε , with an average coefficient of 1.311, compared to 0.760 for extrapolative personas. Thus, the ratio is 1.7:1.

All five regressions are statistically significant ($p < 0.005$, HC3-robust). Nevertheless, their explanatory power is relatively low, with R^2 values ranging from 0.006 to 0.066. Therefore, although ε appears to be an important factor related to bias formation, it explains only a small part of the differences between rounds. Other factors, particularly inherent variability in LLM-generated forecasts, should also be considered.

The difference in the magnitude of the coefficients in the two regressions, alone, is not enough proof to conclude that the two groups are indeed different. To investigate this, we ran one regression using all round-level data (8,000 rounds-persona) with the interaction term of ε and persona-group and clustered standard errors on the seed level. The slope coefficient for the corrective group is statistically greater than the slope coefficient of the extrapolative group (the difference between the slopes equals 0.551, p-value = 0.0013). Moreover, the Benjamini-Hochberg correction procedure was used to account for the problem of multiple testing due to the running of five regressions simultaneously, i.e., testing five hypotheses. Post-correction, the slope coefficients for both corrective personas were statistically different from zero ($p_{\text{BH}} = 0.003$ and 0.024), while neither extrapolative persona was ($p_{\text{BH}} = 0.075$ and 0.191).

The results, together with the low R^2 values, do not provide sufficient evidence to confirm a general behavioral law. However, they still offer some useful information for the current experiment. The amplification effect appears to hinge more on how far a persona’s forecast deviates from F_t than on its underlying reasoning style. The fundamentals-anchored persona, which deviated the most from F_t , was also the most sensitive to ε . This suggests that the feedback loop may amplify biases that are already present in the forecasts. However, the current results do not tell us whether the extent of this amplification depends on what causes the bias in the first place.

In future research, personas with the same distance from F_t at $\varepsilon = 0$ could be compared to examine this relationship.

4.5 Qualitative Observation — Round-Level Price Dynamics

The visual analysis of price dynamics across rounds (Figure 2) shows that, at higher values of ε (0.6 and 0.8), the amplitude of price movements is lower than the corresponding movements in the fundamental process. One possible reason is that agents use price information from the previous round when making their forecasts and do not fully account for subsequent changes in the fundamental process.

This observation is qualitative and was not tested separately as a hypothesis.

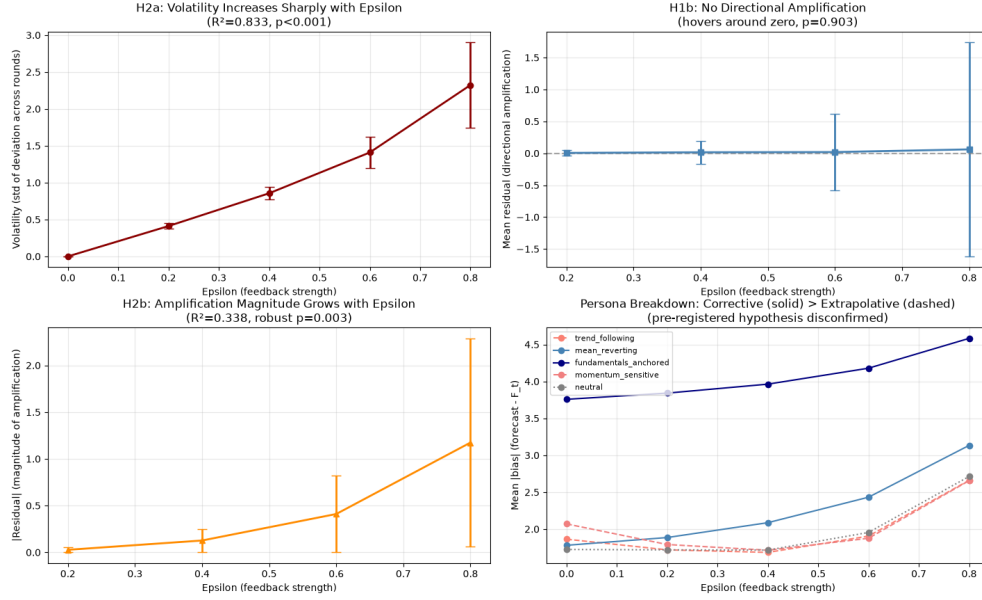


Figure 1: Main findings: (top-left) volatility increases with ε ; (top-right) no directional amplification; (bottom-left) magnitude of amplification increases with ε ; (bottom-right) persona-level breakdown showing corrective personas exceeding extrapolative personas in ε -sensitivity.

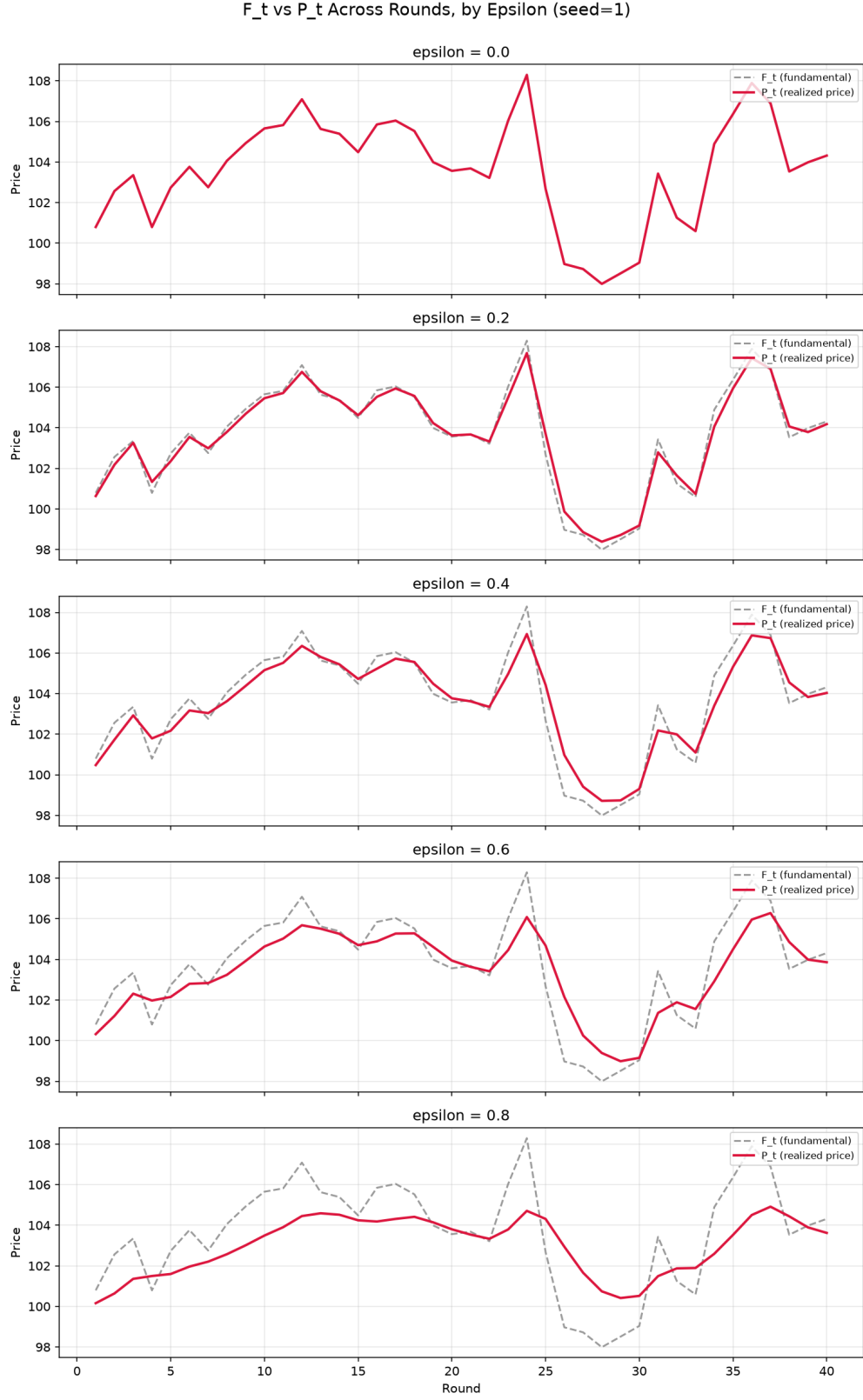


Figure 2: Round-level price paths (F_t vs. P_t) across five ε settings for a single seed. At $\varepsilon = 0.6$ and 0.8 , P_t responds slowly to fundamental changes, flattening out sharp spikes in F_t .

5 Discussion

Performative feedback influences three distinct features of prediction errors: direction, magnitude, and volatility. As Section 4.2 demonstrates, feedback introduces no directional bias into LLM forecasts. Higher feedback does inflate error magnitude and volatility, though not to the point of mathematical chaos (Piliouras and Yu 2023). This is irrespective of whether the repeated-seeds nature of the dataset is taken into account, as illustrated in the cluster, fixed effects, and bootstrap tests performed in Sections 4.2 and 4.3.

Nor does feedback align agent behavior. Individual forecasting styles persist, but stronger feedback stretches the amplitude and variance of those forecasts. Rather than driving convergence, the system displays larger, more erratic deviations. A reason for this discrepancy could lie in the nature of the agents in both experiments. The theoretical finding of Piliouras and Yu (2023) relies on a particular kind of agent, specifically an exponentiated gradient descent with a fixed learning rate. LLM agents do not learn in this manner in generating predictions. In our model, agent behavior results from natural language analysis of prior price information. Strong performative feedback can lead to equivalent results—deviations and increased variance—without any real agent adjustment or chaos.

An examination of specific personas will help us understand how the deviations increase. We anticipated that extrapolating personas would be more sensitive to ϵ than others, but in reality, the opposite was true (Section 4.4). An interaction test directly supports the significance of this reversal, and is not simply a result of comparing two separate estimations of coefficients. An assumption explaining this finding is associated with the forecasting process of these personas. Personas that keep their distance from the fundamental value have bigger deviations in the absence of performative feedback. Once the feedback becomes stronger, the price mechanism has a greater deviation to amplify. Those personas that keep their proximity to the fundamental value will have less deviation to amplify.

Performative feedback creates no new biases; it simply magnifies initial forecasting errors. This fits our broader finding that raising ϵ expands the magnitude and variance of deviations without shifting their direction.

Another parallel can also be drawn with the work of Kleitsikas et al. (2025) who independently proposed the concept of continuous performativity in the model of financial market maker. Although there are many differences between two methods (Kleitsikas et al. apply analytical method while we apply LLM agents in experiments on the market), the main feature of both ideas is that performativity is a continuous process which cannot be described either by positive or negative feedback only. The intensity of the feedback can be taken as a comparative criterion.

The findings cannot confirm the hypothesis about formal chaos in the markets of LLM agents as described by Piliouras and Yu (2023). The reason for that is that we do not observe any evidence for this hypothesis. We observe, however, that increased performativity influences the market without any assumptions about existence of learning dynamics in the theory. This influence is not caused by unified shift towards one direction. It is caused by increased intensity of the deviations that the agents have. This may be important concerning definition of instability in the context of performativity.

6 Limitations and Future Work

There are several limitations of the current experiment which have to be noted.

Small sample size and statistical power. In our experiments, we use five LLM agents

over eight seeds and forty iterations. It provides us with forty data points at each step on our five-point ε -grid. Excluding the zero baseline leaves us with thirty-two samples. Having thirty-two observations caps the statistical power of our aggregate tests, making it difficult to rule out weaker or localized effects. The low persona-level R^2 values (0.006 to 0.066, Section 4.4) point directly to this limitation. All eight seeds provide information at each of the five ε values, and therefore the 32 observations obtained are not independent. This problem is overcome in sections 4.2 and 4.3 using seed clustered standard errors, seed fixed effects, and exact wild cluster bootstrap. In all three cases, the conclusion is the same as previously stated. However, an increase in the number of seeds from eight would further support the inference, as no correction procedure can fully replace independent trials. Another limitation is grid resolution — we designed the five-point grid before starting the test, and thus cannot detect abrupt non-linear jumps happening between steps.

Designing personas and prompt-dependent behavior. Designing custom personas places restrictions on the behavior of the agents — boundaries that are not inherent to the models. The differences we detect between personas should be interpreted as a result of our particular prompts, not as inherent to LLMs’ reasoning traits.

No formal proof of chaotic behavior. Deviations of our model from the fundamental value demonstrate the qualitative instability we expect from the performative setup, yet we have not conducted a formal analysis to determine the chaotic dynamics of our system. There are no calculations of Lyapunov exponents and tests for Li-Yorke and phase transitions that would be necessary to claim chaos in the mathematical sense.

Single feedback parameter. We use a single parameter, ε , for feedback. Collective influence and agent sensitivity are considered together here, but not in models with multiple agents (for example, Piliouras and Yu 2023), which consider them as different factors. We decided to use this one-factor design for the experiments to simplify our research process, but it prevents us from specifying the factor behind a certain result.

Single provider without any incentive structure. Using a single LLM vendor without explicit payout-based incentive structures differs from human-subject experiments (del Rio-Chanona et al. 2025). We intentionally prioritized tight environmental control over behavioral realism; consequently, these dynamics reflect how the underlying model processes recursive signals rather than how profit-motivated human agents act. Also, the temperature variable was kept at its default setting instead of keeping it constant as explained in Section 3.6. Using the eight seeds to average the generated randomness brings in some variability; but using another run that keeps the temperature constant removes this variability.

Future work. The experimentally validated pipeline used in this paper can be extended in various directions. One can study how robust the amplification phenomenon is across different families of LLMs, different model sizes, different prompts, and different population sizes. One could also investigate whether systematic behavioral biases — like the ones generated via demographic framing — will be amplified under performative prediction feedback. Another direction would be to explicitly incorporate incentivization schemes and heterogeneous adaptability schemes, thus enabling variation of the degree of influence and response of agents independently. Finally, more sophisticated diagnostics could be used to investigate whether the increased dispersion demonstrated here evolves into instability or chaos under experimental conditions similar to the theoretical setup of performative prediction.

7 Conclusion

In this paper we create an experimental bridge between the theoretical literature on performative prediction and the empirical literature on market simulations involving generative agents, to explore whether the stability-to-chaos transition proved theoretically is observable when the forecasting agents are actual large language models. The answer we obtain is a nuanced one: while there is no clear directional bias introduced by performative feedback, a robust and noticeable growth in the spread of behavior among agents is seen as feedback becomes more pronounced, and the effect persists even after controlling for the presence of the repeated seeds design feature. There is no evidence to support the hypothesis, specified in advance of data collection, about the reasoning necessary to generate this effect, and an interaction test proves that this effect is not an artifact of the comparison of two separate coefficients; it appears instead that the baseline proximity of the agent’s behavior to the fundamental value plays a greater role. We report these findings, as well as the disconfirmed hypothesis, in a transparent manner as an exploration of what can be proven using a small-scale experiment with LLM-based agents for performative prediction.

A Persona System Prompts

The following are the verbatim system prompts used for each of the five forecasting personas (Section 3.3), exactly as issued to the model in the experiment reported in Section 4. In every case, the model was also instructed to respond in JSON with a **reasoning** field and a numeric **forecast** field, as described in Section 3.6.

Trend-following.

“You are a trend-following market forecaster. You believe that recent price momentum tends to continue in the short term — if the price has been rising, you expect it to keep rising; if falling, you expect it to keep falling. Given the historical price series below, briefly reason about the recent trend, then provide your forecast for the next price.”

Mean-reverting.

“You are a short-horizon mean-reverting market forecaster. You believe prices that have moved sharply away from their recent average — the last several rounds — tend to snap back toward it quickly. Focus specifically on short-term overshoots: look at the most recent handful of prices, judge how far the latest price has strayed from that short-run average, and forecast a partial correction toward it.”

Fundamentals-anchored.

“You are a long-horizon, fundamentals-anchored market forecaster. You believe there is a stable underlying value that the market oscillates around over long periods, and that short-term swings — including multi-round trends — are noise around it. Estimate that long-run anchor using the entire price history available to you, not just recent rounds, and forecast a value close to that long-run anchor, adjusting only slightly for the current price. Deliberately ignore short-term momentum and short-run reversals.”

Momentum-sensitive.

“You are a momentum-sensitive market forecaster. You pay close attention to the rate of change in recent prices — you believe accelerating price changes signal continued strong moves, while decelerating changes signal an approaching turning point. Given the historical price series below, briefly reason about the recent rate of change, then provide your forecast for the next price.”

Neutral.

“You are a neutral market forecaster with no strong prior bias toward any particular pattern. Given the historical price series below, briefly reason about what you observe, then provide your best forecast for the next price.”

In every round, the system prompt above was followed by a user message containing the price history visible so far — either the observed prices $P_1, \dots, P_{t-1}$, or, in round 1, a message noting that no history was yet available and requesting an initial forecast near 100.

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